Test Layer 3 Cross-Connect Traffic: test_l3.py

2025-09-15-22-15-38

Objective

The Layer 3 Traffic Generation Test is designed to test the performance of the Access Point by running layer 3 Cross-Connect Traffic. Layer-3 Cross-Connects represent a stream of data flowing through the system under test. A Cross-Connect (CX) is composed of two Endpoints, each of which is associated with a particular Port (physical or virtual interface).

Device Under Test Information

Device Under Test	DUT Name	
	DUT Hardware Version	
	DUT Software Version	
	DUT Serial Number	

Test Configuration

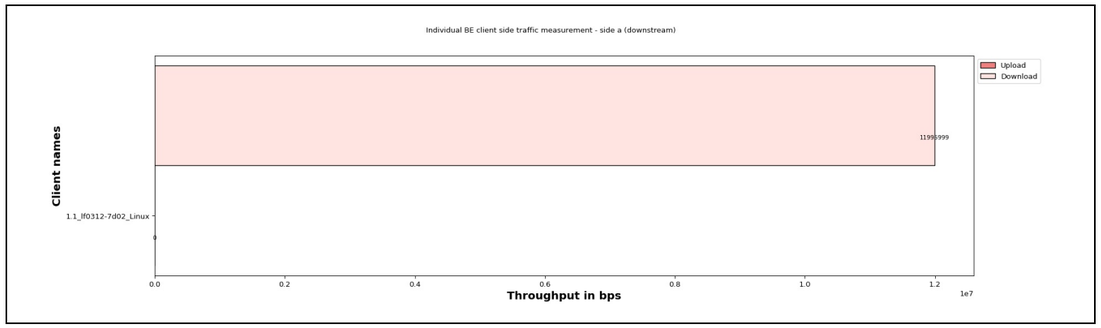
Test Configuration	LANforge ip	192.168.204.75
	LANforge port	8080
	Upstream	1.1.eth1
	Test Duration	30s
	Polling Interval	1s
	Total No. of Devices	1

Radio Configuration

1.1.wiphy0	SSID	NETGEAR_2G_wpa2
	Security	wpa2
	Wifi mode set	AUTO
	Wifi Enable Flags	['wpa2_enable', '80211u_enable', 'create_admin_down']

Individual throughput mcast download bps: 12000000 traffic BE (WiFi).

The below graph represents individual throughput for 1 clients running BE (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

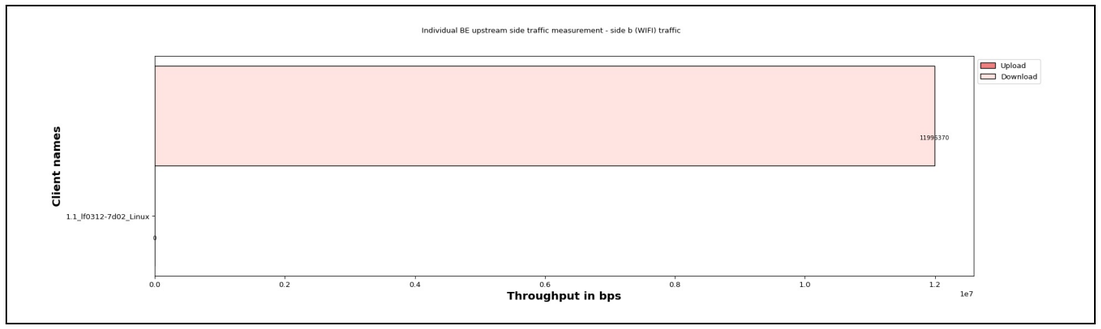


Client Alias	Host eid	Host Name	Device Type / Hw Ver	Endp Name	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Upload Rate Per Client	Offered Download Rate Per Client	Upload Rate Per Client	Download Rate Per Client	Drop Percentage (%)
				LT-												

1.1_lf0312-7d02_Linux	1.1	lf0312-7d02	Linux/x86-64	sta0000-0-BE-A	1.1.sta0000	802.11bgn-AC 20 2x2	04:f0:21:89:2f:f4	NETGEAR_2G_wpa2	11	BE	LF/UDP	0	12000000	0	11995999	0.006302

Individual throughput upstream endp, offered upload bps: 0 offered download bps: 12000000 /station for traffic BE (WiFi).

The below graph represents individual throughput for 1 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



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1.1_lf0312-7d02_Linux	1.1	lf0312-7d02	Linux/x86-64	LT-sta0000-0-BE-B	1.1.sta0000	802.11bgn-AC 20 2x2	04:f0:21:89:2f:f4	NETGEAR_2G_wpa2	11	BE	LF/UDP	12000000	0	0	11996370	0.0

Layer 3 Cx Traffic Last Reporting Interval 1.1.eth1

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954854	09_15_2025_22_17_34	1	0	0	12000000	12000000	MTU	MTU	MTU	MTU	-1	1.1.eth1	55947	5037026	1 Gbps	1 Gbps	NaN	NaN	NaN	00:03:2d:3f:7d:03	[channel: 1]	0	0	0	0	0	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.1.sta0000

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1757954854	09_15_2025_22_17_34	1	0	0	12000000	12000000	MTU	MTU	MTU	MTU	-1	1.1.sta0000	3023112	406	115.6 Mbps	104 Mbps	-53 dBm	94:A6:7E:74:26:22	802.11bgn-AC 20 2x2	04:f0:21:89:2f:f4	11	1	0	0	0	0	0.0	11959420	12399436	30971	0.0